

MTH102

Linear Algebra

2025–2026–I SEMESTER

PROBLEM SETS

17 October 2025

Chapter 1. Problem set 1

Exercise 1.1. Determine the elements of the following sets.

- $\{x \in \mathbb{N} : x^2 - 1 = 0\}$.
- $\{x \in \mathbb{Z} : x^2 - 1 = 0\}$.

Exercise 1.2. What does it mean to say that A is not a subset of B ?

Exercise 1.3. Show that if A is a subset of B and B is a subset of C , then A is a subset of C .

Exercise 1.4. Show that no element n of $\mathbb{N}$ satisfies $n^4 - 5n^2 + 6 = 0$.

Exercise 1.5. What does it mean to say that two sets A, B are not equal?

Exercise 1.6. Consider the sets

$$\{1, 2, 3, 4\}, \{3, 4, 5, 6\}.$$

Determine the elements common to these sets.

Exercise 1.7. Determine the union of the sets

$$\{1, 2, 3, 4\}, \{1, 3, 5, 7\}.$$

Exercise 1.8. If A, B are sets, show that

$$A \subseteq A \cup B, B \subseteq A \cup B.$$

Exercise 1.9. If A, B are subsets of a set C , then $A \cup B$ is a subset of C .

Exercise 1.10. If A is a set, show that

$$A \cup A = A.$$

Exercise 1.11 (Commutative property). If A, B are sets, show that

$$A \cup B = B \cup A.$$

Exercise 1.12 (Associative property). If A, B, C are sets, show that

$$A \cup B \cup C = (A \cup B) \cup C = A \cup (B \cup C).$$

Exercise 1.13. Determine the intersection of the sets

$$\{1, 2, 3, 4\}, \{1, 3, 5, 7\}.$$

Exercise 1.14. If A, B are sets, show that

$$A \cap B \subseteq A, A \cap B \subseteq B.$$

Exercise 1.15. If A, B, C are sets satisfying

$$C \subseteq A, C \subseteq B,$$

then show that

$$C \subseteq A \cap B$$

holds.

Exercise 1.16. Identify the integers among

$$1, 2, 3, 4, \dots, 20$$

which are divisible by at least one of 2 and 5.

Exercise 1.17 (Inclusion-exclusion principle). Show that for finite subsets A, B, C of a set X ,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |C \cap A| + |A \cap B \cap C|.$$

Exercise 1.18. Determine the number of integers among

$$1, 2, 3, 4, \dots, 100,$$

which are divisible by at least one of 6, 10, 15.

Exercise 1.19. If A is a set, show that

$$A \cap A = A.$$

Exercise 1.20 (Commutative property). If A, B are sets, show that

$$A \cap B = B \cap A.$$

Exercise 1.21 (Associative property). If A, B, C are sets, show that

$$A \cap B \cap C = (A \cap B) \cap C = A \cap (B \cap C).$$

Exercise 1.22. Show that

$$A^c = \{x \in X : x \notin A\},$$

where X denotes the underlying universal set.

Exercise 1.23. Determine the complement of $\{1, 2, 3, 4\}$ in $\{1, 3, 5, 7\}$, and the complement of $\{1, 3, 5, 7\}$ in $\{1, 2, 3, 4\}$.

Exercise 1.24. If A, B are subsets of a set X , show that

$$A^c \cap B = B \setminus A.$$

Exercise 1.25. If A is a subset of a set X , then show that

$$A \cup A^c = X, A \cap A^c = \emptyset, (A^c)^c = A.$$

Exercise 1.26. If A, B are subsets of a set X , then show that

$$(A \cup B)^c = A^c \cap B^c, (A \cap B)^c = A^c \cup B^c.$$

Exercise 1.27. Let A, B be subsets of a set X . Show that the following statements are equivalent.

$$\begin{aligned} A &\subseteq B, \\ A \cap B &= A, \\ A \cup B &= B, \\ B^c &\subseteq A^c. \end{aligned}$$

Exercise 1.28. Let A, B be subsets of a set X . Show that the sets

$$A \setminus B, B \setminus A$$

are disjoint.

Exercise 1.29. Determine the symmetric difference of $\{1, 2, 3, 4\}$ and $\{1, 3, 5, 7\}$.

Exercise 1.30. If A, B are sets, then show that

$$\begin{aligned} A \Delta B &= (A \setminus B) \cup (B \setminus A), \\ A \Delta B &= B \Delta A, \\ A \cup B &= (A \Delta B) \cup (A \cap B), \\ (A \Delta B) \cap (A \cap B) &= \emptyset. \end{aligned}$$

Exercise 1.31. Write down the power set of each of the following sets:

$$\{1, 2\}, \{1, 2, 3\}, \{1, \{2, 3\}\}, \{1, 2, 3, 4\}.$$

Exercise 1.32. If A, B are finite sets, then show that

$$|A \times B| = |A| \times |B|,$$

where for a set X , its number of elements is denoted by $|X|$.

Exercise 1.33. Determine the cartesian product of $[1, 2]$ and $[3, 4] \cup [5, 6]$.

Exercise 1.34. Identify the set

$$\left\{x \in \mathbb{R} \setminus \{0\} : x + \frac{1}{x} \geq 2\right\}.$$

Exercise 1.35. Let A, B be subsets of $\mathbb{R}$ defined by

$$\begin{aligned} A &= \{x \in \mathbb{R} : x^2 \geq 0\}, \\ B &= \{x \in \mathbb{R} : x^3 \geq 0\}. \end{aligned}$$

Determine the sets $A \cup B, A \cap B, A \setminus B, B \setminus A$.

Exercise 1.36. For a positive integer k , let A_k denote the set of integral multiples of k , that is,

$$A_k := \{r \in \mathbb{Z} : r = k\ell \text{ for some } \ell \in \mathbb{Z}\}.$$

Let m, n be positive integers. For each of the following statements, determine the equivalent conditions on the integers m, n .

1. $A_m \subseteq A_n$
2. $A_m \subsetneq A_n$
3. $A_m \not\subseteq A_n$
4. $A_m = A_n$
5. $A_m \cap A_n = \emptyset$
6. $A_m \setminus A_n \neq \emptyset$
7. $A_m \setminus A_n \neq \emptyset$ or $A_n \setminus A_m \neq \emptyset$
8. $A_m \setminus A_n \neq \emptyset$ and $A_n \setminus A_m \neq \emptyset$

Exercise 1.37. Show that $\mathbb{R}$ is a subset of $\mathbb{C}$.

Exercise 1.38. If z, w are complex numbers, then show that

1. $\overline{z + w} = \overline{z} + \overline{w}$,
2. $\overline{z \cdot w} = \overline{z} \cdot \overline{w}$,
3. $|z| \geq 0$, and $|z| = 0$ if and only if $z = 0$,
4. $|z \cdot w| = |z| |w|$.

Exercise 1.39. If z is a complex number, then show that

$$z \cdot \overline{z} = |z|^2.$$

Exercise 1.40. For any real number θ , show that

$$|\cos \theta + i \sin \theta| = 1.$$

Chapter 2. Induction principles

Exercise 2.1. If A is a nonempty subset of $\mathbb{N}$, and a_1, a_2 are least elements of A , then show that $a_1 = a_2$.

Exercise 2.2. Show that

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

for all $n \in \mathbb{N}$.

Exercise 2.3. Show that

$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \cdots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

for all $n \in \mathbb{N}$.

Exercise 2.4. Show that

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \cdots + \frac{1}{n(n+1)(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$$

for all $n \in \mathbb{N}$.

Exercise 2.5. Show that

$$3 + 11 + \cdots + (8n-5) = 4n^2 - n$$

for all $n \in \mathbb{N}$.

Exercise 2.6. Show that

$$1^2 + 3^2 + \cdots + (2n-1)^2 = \frac{4n^3 - n}{3}$$

for all $n \in \mathbb{N}$.

Exercise 2.7. Show that

$$1^2 - 2^2 + 3^2 - 4^2 + \cdots + (-1)^{n+1}n^2 = (-1)^{n+1} \frac{n(n+1)}{2}$$

for all $n \in \mathbb{N}$.

Exercise 2.8. Show that

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6},$$

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

hold for any positive integer n .

Exercise 2.9. Show that

$$\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{2n-1} - \frac{1}{2n}$$

for all $n \in \mathbb{N}$.

Exercise 2.10. Show that $n^3 + 5n$ is divisible by 6 for all $n \in \mathbb{N}$.

Exercise 2.11. Show that $5^{2n} - 1$ is divisible by 8 for all $n \in \mathbb{N}$.

Exercise 2.12. Show that $5^n - 4n - 1$ is divisible by 16 for all $n \in \mathbb{N}$.

Exercise 2.13. Show that $6^n - 5n - 1$ is divisible by 25 for all $n \in \mathbb{N}$.

Exercise 2.14. Show that $n^3 + (n+1)^3 + (n+2)^3$ is divisible by 9 for all $n \in \mathbb{N}$.

Exercise 2.15. Show that

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

for all $n \in \mathbb{N}$ satisfying $n > 1$.

Exercise 2.16. Show that $3^n \geq n^2$ for all $n \in \mathbb{N}$.

Exercise 2.17. Show that $n! > 2^n$ for all $n \in \mathbb{N}$ satisfying $n \geq 4$.

Here are a few exercises where strong induction is useful.

Exercise 2.18. Show that every positive integer greater than 1 is a prime or is a product of prime numbers.

Exercise 2.19. Let the integers $x_1, x_2, \dots$ be defined by

$$x_1 := 1,$$

$$x_2 := 2,$$

$$x_{n+2} := \frac{1}{2}(x_n + x_{n+1})$$

for all $n \in \mathbb{N}$. Show that $1 \leq x_n \leq 2$ for all $n \in \mathbb{N}$.

Exercise 2.20. The Fibonacci sequence $F_0, F_1, F_2, \dots$ is defined by

$$F_0 := 0,$$

$$F_1 := 1,$$

$$F_2 := 1,$$

$$F_{n+2} := F_n + F_{n+1}$$

for all $n \in \mathbb{N}$. Show that

$$F_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right)$$

for all $n \in \mathbb{N} \cup \{0\}$.

Exercise 2.21. Show that

$$F_0^2 + F_1^2 + F_2^2 + \cdots + F_n^2 = F_n F_{n+1}$$

for all $n \in \mathbb{N} \cup \{0\}$.

Chapter 3. Matrices

Exercise 3.1. If A, B, C are matrices of the same size, show that

$$A + (B + C) = (A + B) + C.$$

Exercise 3.2. If A, B are matrices of the same size, show that

$$A + B = B + A.$$

Exercise 3.3. If A is an $m \times n$ matrix, show that

$$A + 0_{m \times n} = 0_{m \times n} + A = A$$

holds, where $0_{m \times n}$ denotes the $m \times n$ zero matrix, that is, the $m \times n$ matrix whose entries are all zero.

Exercise 3.4. Show that if A is a matrix and c, d are real numbers, then

$$(c + d)A = cA + dA$$

and

$$c(dA) = (cd)A.$$

Exercise 3.5. If A is a matrix, what is $(A^T)^T$?

Exercise 3.6. Show that if A, B are matrices of the same size, then

$$(A + B)^T = A^T + B^T.$$

Exercise 3.7. Show that if A denotes a matrix, then

$$(A^T)^T = A.$$

Exercise 3.8. Show that if A, B, C are matrices such that the products $A(BC)$ and $(AB)C$ are defined, then

$$A(BC) = (AB)C.$$

Exercise 3.9. Provide examples to show that in general, matrix multiplication is not commutative, that is, $AB \neq BA$ for some matrices A, B .

Exercise 3.10 ().** Show that for any $n \in \mathbb{N}$ with $n \geq 2$, there are $n \times n$ matrices A, B such that

$$AB \neq BA.$$

? Question

Can induction be used for the above exercise?

Exercise 3.11. If A is an $n \times n$ matrix, show that

$$AI_n = I_n A = A$$

holds, where I_n denotes the $n \times n$ diagonal matrix, with all diagonal entries equal to 1, that is,

$$I_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}.$$

Exercise 3.12. If A is an $m \times n$ matrix, show that

$$I_m A = A, A I_n = A$$

where I_m (resp. I_n) denote the $m \times m$ (resp. $n \times n$) identity matrix.

Exercise 3.13. If A is an $m \times n$ matrix with entries in $\mathbb{R}$ and c is a real number, then

$$cA = (cI_n)A = A(cI_m),$$

where I_n (resp. I_m) denotes the $n \times n$ (resp. $m \times m$) identity matrix.

Exercise 3.14. Show that if A, B are matrices, then

$$(AB)^T = B^T A^T.$$

Exercise 3.15. Show that if

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix},$$

then

$$A^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}$$

for all $n \in \mathbb{N}$ with $n \geq 1$, where F_n denotes the n -th Fibonacci number.

Exercise 3.16. Compute

$$\begin{pmatrix} 0 & \dots & 0 & 1 \\ 0 & \dots & 1 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 1 & \dots & 0 & 0 \end{pmatrix}^2.$$

Exercise 3.17. Show that if A, B, C are matrices such that the sum $B + C$, and the products $AB, AC, A(B + C)$ are defined, then

$$A(B + C) = AB + AC.$$

Exercise 3.18. Show that if A, B, C are matrices such that the sum $B + C$, and the products $AB, AC, A(B + C)$ are defined, then

$$(A + B)C = AC + BC.$$

Exercise 3.19. If A is an $m \times n$ matrix, then show that for all $i = 1, 2, \dots, n$, the product Ae_i is equal to the i -th column of A , where e_i denotes the i -th standard basis vector of $\mathbb{R}^n$, that is,

$$e_i = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$$

with 1 in the i -th position, and 0's elsewhere.

Exercise 3.20. Determine whether the identity matrix I_n is invertible, and if it is, find its inverse.

Exercise 3.21. If A is an invertible square matrix, then show that A^{-1} is also invertible, and that

$$(A^{-1})^{-1} = A.$$

Exercise 3.22. Let A, B be invertible $n \times n$ matrices. Show that AB is invertible, and that

$$(AB)^{-1} = B^{-1}A^{-1}.$$

Compare the above exercise with [Exercise 3.14](#).

Exercise 3.23. If A, B are 2×2 matrices, then show that

$$\det(AB) = \det(A) \det(B).$$

Exercise 3.24. Solve the system of equations

$$\begin{aligned} 5x + 2y &= 11, \\ 3x + 4y &= 8. \end{aligned}$$

Exercise 3.25. Solve the system of equations

$$\begin{aligned} 12x - 25y &= -47, \\ -7x + 30y &= 51. \end{aligned}$$

Exercise 3.26. Solve the system of equations

$$\begin{aligned} 2x + 3y + z &= 1, \\ 4x + y + 2z &= 2, \\ 3x + 2y + 3z &= 3. \end{aligned}$$

Exercise 3.27. Let A, B be 2×2 matrices. Show that if AB is invertible, then both A and B are invertible.

Compare the above with [Exercise 3.63](#), [Exercise 3.64](#), [Exercise 3.75](#).

Exercise 3.28. Let A be an $n \times n$ matrix. If P is an invertible $n \times n$ matrix, then show that

$$(PAP^{-1})^k = PA^kP^{-1}$$

holds for any positive integer k .

Exercise 3.29. Let A be a 2×2 matrix. Let V be the set of solutions to the system of equations

$$A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

that is,

$$V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : A \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}.$$

Show that for any $u, v \in V$ and $c \in \mathbb{R}$, the element $u + v$ of $\mathbb{R}^2$ lies in V , and the element cu also lies in V .

Exercise 3.30. Let A be a 2×2 matrix. Let V be the set of vectors which can be expressed as a linear combination of the columns of A , that is¹,

$$V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : \begin{pmatrix} x \\ y \end{pmatrix} = A \begin{pmatrix} s \\ t \end{pmatrix} \text{ for some } s, t \in \mathbb{R} \right\}.$$

Show that for any $u, v \in V$ and $c \in \mathbb{R}$, the element $u + v$ of $\mathbb{R}^2$ lies in V , and the element cu also lies in V .

Compare the following exercise with [Exercise 3.57](#).

Exercise 3.31. Let A be an $m \times n$ matrix. Let V be the set of solutions to the system of equations

$$Au = 0,$$

that is,

$$V = \{v \in \mathbb{R}^n : Av = 0\}.$$

Show that for any $u, v \in V$ and $c, d \in \mathbb{R}$, the element $cu + dv$ of $\mathbb{R}^m$ lies in V .

Compare the following exercise with [Exercise 3.58](#).

Exercise 3.32. Let A be an $m \times n$ matrix. Let V be the set of vectors which can be expressed as a linear combination of the columns of A , that is,

$$V = \{v \in \mathbb{R}^m : v = Au \text{ for some } u \in \mathbb{R}^n\}.$$

Show that for any $u, v \in V$ and $c, d \in \mathbb{R}$, the element $cu + dv$ of $\mathbb{R}^m$ lies in V .

Exercise 3.33. Let A be a square matrix. Show that if A is symmetric and skew-symmetric, then A is the zero matrix.

Exercise 3.34. Show that for any $n \geq 2$, there are infinitely many $n \times n$ orthogonal matrices.

Exercise 3.35. Let A be an orthogonal matrix. Show that A is invertible and that

$$A^{-1} = A^T.$$

Exercise 3.36. Let A, B be orthogonal matrices of the same size. Show that the matrix product AB is also an orthogonal matrix.

Exercise 3.37. Let A, B be square matrices with complex entries. Show that

$$\overline{AB} = \overline{A} \overline{B}.$$

¹Observe that

$$A \begin{pmatrix} s \\ t \end{pmatrix} = sC_1 + tC_2,$$

where C_1, C_2 denote the first and second columns of A respectively.

Exercise 3.38. Let A be a square matrix with complex entries. Show that

$$\overline{A^T} = (\overline{A})^T.$$

Exercise 3.39. Let A, B be square matrices of the same size with complex entries. Show that

$$(A + B)^* = A^* + B^*.$$

Exercise 3.40. Let A be a matrix with complex entries. Show that

$$(A^*)^* = A.$$

Exercise 3.41. Let A, B be matrices with complex entries. Show that

$$(AB)^* = B^* A^*.$$

Exercise 3.42. Let A be an $n \times n$ hermitian matrix. Show that all the diagonal entries of A are real numbers. More generally, show that for any $1 \leq i, j \leq n$, the (i, j) -entry and the (j, i) -entry of A are complex conjugates of each other.

Exercise 3.43. Let A, B be hermitian matrices of the same size. Show that the matrix $A + B$ is also a hermitian matrix. Is the matrix AB also a hermitian matrix?

Exercise 3.44. State and prove an analogue of [Exercise 3.42](#) for skew-hermitian matrices.

Exercise 3.45. Let A, B be skew-hermitian matrices of the same size. Show that the matrix $A + B$ is also a skew-hermitian matrix. Is the matrix AB also a skew-hermitian matrix?

Exercise 3.46. Let A be a square matrix with complex entries. Show that if A is hermitian and skew-hermitian, then A is the zero matrix.

Exercise 3.47. Let A be a unitary matrix. Show that A is invertible and that

$$A^{-1} = A^*.$$

Exercise 3.48. Let A, B be unitary matrices of the same size. Show that the matrix product AB is also a unitary matrix.

Exercise 3.49. Show that any orthogonal matrix is a unitary matrix. Is the converse true?

Exercise 3.50. Determine whether the following matrices are unitary, hermitian, both, or neither.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Exercise 3.51. Let z, w be complex numbers such that $|z|^2 + |w|^2 = 1$. Show that

$$\begin{pmatrix} z & -\overline{w} \\ w & \overline{z} \end{pmatrix}$$

is a unitary matrix.

Exercise 3.52. Let A be a diagonal matrix with complex entries. Show that A is a unitary matrix if and only if all the diagonal entries of A have absolute value equal to 1.

Exercise 3.53. Show that any unitary matrix is a normal matrix. Is the converse true?

Exercise 3.54. Show that any hermitian matrix is a normal matrix. Is the converse true?

Exercise 3.55. Show that any skew-hermitian matrix is a normal matrix. Is the converse true?

Exercise 3.56. Let e_{ij}, e_{kl} be matrix units of the same size. Show that $e_{ij}e_{kl}$ is equal to the matrix unit e_{il} if $j = k$, and is equal to the zero matrix if $j \neq k$, that is,

$$e_{ij}e_{kl} = \begin{cases} e_{il} & \text{if } j = k, \\ 0 & \text{if } j \neq k. \end{cases}$$

Denote the set of $m \times n$ matrices with entries from $\mathbb{R}$ by $M_{m,n}(\mathbb{R})$.

Compare the following exercise with [Exercise 3.31](#).

Exercise 3.57. Let A be an $m \times n$ matrix. Let V be the set of solutions to the system of equations

$$uA = 0,$$

that is,

$$V = \{v \in M_{1,m}(\mathbb{R}) : vA = 0\}.$$

Show that for any $u, v \in V$ and $c, d \in \mathbb{R}$, the element $cu + dv$ of $\mathbb{R}^m$ lies in V .

Compare the following exercise with [Exercise 3.32](#).

Exercise 3.58. Let A be an $m \times n$ matrix. Let V be the set of row vectors which can be expressed as a linear combination of the rows of A , that is,

$$V = \{v \in M_{1,n}(\mathbb{R}) : v = u^T A \text{ for some } u \in \mathbb{R}^m\}.$$

Show that for any $u, v \in V$ and $c, d \in \mathbb{R}$, the element $cu + dv$ of $M_{1,n}(\mathbb{R})$ lies in V .

Exercise 3.59. Solve the system of linear equations in five variables x_1, x_2, x_3, x_4, x_5 :

$$\begin{aligned} x_1 + 2x_2 - x_3 + 4x_4 + x_5 &= 7, \\ 2x_1 + 3x_2 + x_3 + 5x_4 + 2x_5 &= 14, \\ -x_1 + 4x_2 - 2x_3 - 3x_4 + x_5 &= -10. \end{aligned} \tag{1}$$

Exercise 3.60. Solve the system of equations in the variables x_1, x_2, x_3 :

$$\begin{aligned} x_1 + 2x_2 - x_3 &= 1, \\ 2x_1 + 3x_2 + x_3 &= 2, \\ -x_1 + 4x_2 - 2x_3 &= -3, \\ 3x_1 - x_2 + 4x_3 &= 4, \\ 5x_1 + 2x_2 + 3x_3 &= 5. \end{aligned}$$

Exercise 3.61. Solve the system of equations in the variables x_1, x_2, x_3, x_4 :

$$\begin{aligned}
x_1 + 2x_2 - x_3 + 4x_4 &= 5, \\
2x_1 + 3x_2 + x_3 + 5x_4 &= 8, \\
-x_1 + 4x_2 - 2x_3 - 3x_4 &= -4, \\
3x_1 - x_2 + 4x_3 + 2x_4 &= 7.
\end{aligned}$$

Exercise 3.62. Solve the system of equations in the variables x_1, x_2, x_3 :

$$\begin{aligned}
x_1 + 2x_2 - 5x_3 &= 20, \\
2x_1 + 5x_2 - 7x_3 &= 33, \\
-x_1 - 2x_2 + 4x_3 &= -17.
\end{aligned}$$

Exercise 3.63. Let A, B be square matrices of the same size. Suppose that A is invertible and $AB = I$ holds. Show that B is invertible and $B = A^{-1}$.

Exercise 3.64. Let A, B be square matrices of the same size. Suppose that A is invertible and $BA = I$ holds. Show that B is invertible and $B = A^{-1}$.

Compare the above two exercises with [Exercise 3.27](#), [Exercise 3.75](#).

Exercise 3.65. Let A, B be matrices, suppose that the number of columns of A is equal to the number of rows of B . Suppose AB has at least two columns. Let C be the matrix obtained from AB by removing the last column of AB . Show that there exists a matrix B' such that $C = AB'$.

Exercise 3.66. Let A, B be matrices, suppose that the number of columns of A is equal to the number of rows of B . Suppose AB has at least two rows. Let C be the matrix obtained from AB by removing the last row of AB . Show that there exists a matrix A' such that $C = A'B$.

Exercise 3.67. Use elementary row operations to reduce the matrix

$$A = \begin{pmatrix} 1 & 2 & -5 \\ 2 & 5 & -7 \\ -1 & -2 & 4 \end{pmatrix}$$

in its row echelon form. Use the row echelon form to determine whether A is invertible. If A is invertible, use elementary matrices corresponding to the performed elementary row operations to find A^{-1} .

Exercise 3.68. Solve the system of equations in the variables x_1, x_2 :

$$\begin{aligned}
2x_1 - 3x_2 &= 7, \\
-4x_1 + 5x_2 &= -11.
\end{aligned}$$

Exercise 3.69. Transform the matrix

$$\begin{pmatrix} 0 & 2 & -3 & 0 & 0 \\ 1 & -1 & 4 & 0 & 5 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 3 & -5 & 0 & 1 \\ 0 & 3 & -5 & -1 & 2 \end{pmatrix}$$

into a matrix in row echelon form using elementary row operations.

Exercise 3.70. Solve the system of equations in the variables x_1, x_2, x_3, x_4 :

$$\begin{aligned} 2x_2 - 3x_3 &= 0, \\ x_1 - x_2 + 4x_3 &= 5, \\ x_4 &= -2, \\ 3x_2 - 5x_3 &= 1, \\ 3x_2 - 5x_3 - x_4 &= 2. \end{aligned}$$

Exercise 3.71. Find the rank of the matrix

$$\begin{pmatrix} 1 & 2 & -1 & 1 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 1 & -\frac{2}{15} \\ 0 & 0 & 1 & -\frac{1}{14} \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Exercise 3.72. If A is an $m \times n$ matrix, then show that

$$\text{rk}(A) \leq \min\{m, n\}.$$

Exercise 3.73. If A is an $n \times n$ matrix, then show that the following statements are equivalent.

1. The matrix A is invertible.
2. The rank of A is n .

Exercise 3.74. If A is an invertible square matrix, then show that A^{-1} is also invertible, and that

$$(A^{-1})^{-1} = A.$$

Exercise 3.75. If A, B are square matrices of the same size satisfying $BA = I$, then show that the matrices A, B are invertible, and are the inverses of each other.

Compare the above with [Exercise 3.63](#), [Exercise 3.64](#), [Exercise 3.27](#).

Exercise 3.76. Solve the system of equations in the variables x, y, z :

$$\begin{aligned} x + 2y + 3z &= 1, \\ 4x + 5y + 6z &= 2, \\ 7x + 8y + 10z &= 3. \end{aligned}$$

Exercise 3.77. Show that the determinant of the identity matrix I_3 is 1, and that

$$\text{adj}(I_3) = I_3.$$

Exercise 3.78. Let

$$A = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix},$$

where a, b, c are scalars. Show that $\det(A) = abc$, and that

$$\operatorname{adj}(A) = \begin{pmatrix} bc & 0 & 0 \\ 0 & ca & 0 \\ 0 & 0 & ab \end{pmatrix}.$$

Exercise 3.79. Let

$$A = \begin{pmatrix} 1 & -5 & 6 \\ -4 & 8 & -9 \\ 7 & 2 & 10 \end{pmatrix}.$$

Find $\det(A)$ and $\operatorname{adj}(A)$.

Exercise 3.80. Let

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 3 & 2 & 1 \\ 4 & 1 & 2 \end{pmatrix}.$$

Find $\det(A)$ and $\operatorname{adj}(A)$.

Exercise 3.81. Let

$$A = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}$$

where a, b are scalars. Show that $\det(A) = ab$, and that

$$\operatorname{adj}(A) = \begin{pmatrix} b & 0 \\ 0 & a \end{pmatrix}.$$

Exercise 3.82. Compute the determinant of the matrix

$$A = \begin{pmatrix} 1 & -5 & 6 \\ 2 & 0 & 9 \\ -1 & 2 & -1 \end{pmatrix}$$

to determine whether it is invertible. If A is invertible, then find its inverse using its adjoint.

Exercise 3.83. Use Gaussian elimination to determine the row echelon form of the matrix

$$A = \begin{pmatrix} 1 & -5 & 6 \\ 2 & 0 & 9 \\ -1 & 2 & -1 \end{pmatrix}.$$

Use the row echelon form to determine whether A is invertible. If A is invertible, then find its inverse using elementary matrices, corresponding to the elementary row operations used to transform A into its row echelon form.

Exercise 3.84. Solve the following system of equations in the variables x, y, z .

$$\begin{aligned} x + 2y + 3z &= 1, \\ 4x - 5y + 6z &= 2, \\ 7x - y + 10z &= 3. \end{aligned}$$

Chapter 4. The space $\mathbb{R}^n$

Exercise 4.1. Let S be a subset of $\mathbb{R}^n$, containing precisely one element v . Show that S is linearly independent if and only if $v \neq 0$.

Exercise 4.2. In each of the following, determine whether the given vectors are linearly independent or dependent. Justify your answers.

1. $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$, and $\begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}$ in $\mathbb{R}^3$.
2. $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, and $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ in $\mathbb{R}^3$.
3. $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$ in $\mathbb{R}^2$.
4. $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ in $\mathbb{R}^2$.
5. $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$, and $\begin{pmatrix} 7 \\ 8 \\ 10 \end{pmatrix}$ in $\mathbb{R}^3$.

Exercise 4.3. Let $v_1 = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$, $v_2 = \begin{pmatrix} 4 \\ 0 \\ -1 \end{pmatrix}$, $v_3 = \begin{pmatrix} -1 \\ 5 \\ 2 \end{pmatrix}$. Determine whether the vectors v_1, v_2, v_3 are linearly independent or dependent. Justify your answer.

Exercise 4.4. Determine whether the vectors $\begin{pmatrix} 1 \\ 2 \\ -1 \\ 3 \end{pmatrix}$, $\begin{pmatrix} 4 \\ 0 \\ 5 \\ -2 \end{pmatrix}$, $\begin{pmatrix} -2 \\ 1 \\ 3 \\ 4 \end{pmatrix}$, and $\begin{pmatrix} 3 \\ -1 \\ 2 \\ 1 \end{pmatrix}$ in $\mathbb{R}^4$ are linearly independent or dependent. Justify your answer.

Exercise 4.5. Show that the standard basis vectors

$$e_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_3 := \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, e_n := \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

of $\mathbb{R}^n$ are linearly independent.

Exercise 4.6. Show that any finite subset of $\mathbb{R}^n$ containing the zero vector is linearly dependent.

Exercise 4.7. Show that any subset of a linearly independent set of vectors is also linearly independent.

Exercise 4.8. Let $v_1, v_2, \dots, v_k$ be elements of $\mathbb{R}^n$. If $k > n$, then the vectors $v_1, v_2, \dots, v_k$ are linearly dependent, that is, at least one of the vectors can be expressed as a linear combination of the others.

Exercise 4.9. Determine the linear subspaces of $\mathbb{R}$.

Exercise 4.10. Let V denote the set of all vectors of $\mathbb{R}^n$ whose first component is equal to zero, that is,

$$V = \left\{ \begin{pmatrix} 0 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix} : x_i \in \mathbb{R} \text{ for all } i = 2, 3, \dots, n \right\}.$$

Show that V is a linear subspace of $\mathbb{R}^n$.

Exercise 4.11. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}$ is a linear subspace of $\mathbb{R}^3$.

Exercise 4.12. Let (a, b, c) be a nonzero element of $\mathbb{R}^3$. Show that the set

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : ax + by + cz = 0 \right\}$$

is a linear subspace of $\mathbb{R}^3$.

Exercise 4.13. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x \geq 0, y \geq 0 \right\}$ is not a linear subspace of $\mathbb{R}^2$.

Exercise 4.14. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x = 1 \right\}$ is not a linear subspace of $\mathbb{R}^2$.

Exercise 4.15. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x = y \right\}$ is a linear subspace of $\mathbb{R}^2$.

Exercise 4.16. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : z = 0 \right\}$ is a linear subspace of $\mathbb{R}^3$.

Exercise 4.17. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 1 \right\}$ is not a linear subspace of $\mathbb{R}^3$.

Exercise 4.18. Show that $\mathbb{R}^n$ is a linear subspace of itself.

Exercise 4.19. Show that the set containing only the zero vector, that is, the set $\{0\}$, is a linear subspace of $\mathbb{R}^n$.

Exercise 4.20. Determine the subspaces of $\mathbb{R}^2$.

Exercise 4.21. Determine the subspaces of $\mathbb{R}^3$.

Exercise 4.22. Let A be an $m \times n$ matrix. Let V be the set of solutions to the system of equations

$$Au = 0,$$

that is,

$$V = \{v \in \mathbb{R}^n : Av = 0\}.$$

Show that V is a linear subspace of $\mathbb{R}^n$.

Exercise 4.23. Let A be an $m \times n$ matrix. Let V be the set of vectors which can be expressed as a linear combination of the columns of A , that is,

$$V = \{v \in \mathbb{R}^m : v = Au \text{ for some } u \in \mathbb{R}^n\}.$$

Show that V is a linear subspace of $\mathbb{R}^m$.

Exercise 4.24. Let A be an $m \times n$ matrix with real entries. Show that for any invertible $m \times m$ matrix E , the null space $\text{null}(A)$ is equal to the null space $\text{null}(EA)$.

Exercise 4.25. Let m, n be positive integers with $m \geq 2$. Provide an example of an $m \times n$ matrix A with real entries and an invertible $m \times m$ matrix E such that the column space $\text{col}(A)$ of A is not equal to the column space $\text{col}(EA)$ of EA .

Exercise 4.26. Let U, V be the subsets of $\mathbb{R}^3$ defined as

$$U = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + 2y = 0 \right\},$$

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}.$$

Show that both U and V are linear subspaces of $\mathbb{R}^3$. Determine the intersection $U \cap V$ of U and V . Is the intersection $U \cap V$ a linear subspace of $\mathbb{R}^3$?

Exercise 4.27. Show that the intersection of two linear subspaces of $\mathbb{R}^n$ is also a linear subspace of $\mathbb{R}^n$.

Exercise 4.28. Show that the linear span of any set of vectors in $\mathbb{R}^n$ is a linear subspace of $\mathbb{R}^n$.

Exercise 4.29. Let V be a linear subspace of $\mathbb{R}^n$ and S be a finite subset of V . Show that the linear span of S is contained in V , that is,

$$\text{span}(S) \subseteq V.$$

Exercise 4.30. Let $v_1, v_2, \dots, v_k \in \mathbb{R}^n$ be linearly independent vectors. Show that if $u \in \mathbb{R}^n$ is not in the span of $\{v_1, v_2, \dots, v_k\}$, then the set of vectors $\{v_1, v_2, \dots, v_k, u\}$ is linearly independent.

Exercise 4.31. Let $v_1, v_2, \dots, v_k$ be elements of $\mathbb{R}^n$. Show that if $u \in \mathbb{R}^n$ is in the span of $\{v_1, v_2, \dots, v_k\}$, then the set of vectors $\{v_1, v_2, \dots, v_k, u\}$ is linearly dependent.

Exercise 4.32. Let U, V be linear subspaces of $\mathbb{R}^n$. Show that the sum $U + V$ is equal to the sum $V + U$.

Exercise 4.33. Let U, V be linear subspaces of $\mathbb{R}^n$. Show that $U \subseteq U + V$ and $V \subseteq U + V$.

Exercise 4.34. Let V be a linear subspace of $\mathbb{R}^n$. Show that $V + V = V$.

Exercise 4.35. Let U, V be the subsets of $\mathbb{R}^3$ defined as

$$U = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + 2y = 0 \right\},$$

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}.$$

Determine the sum of U and V .

Exercise 4.36. Show that the sum of two linear subspaces of $\mathbb{R}^n$ is also a linear subspace of $\mathbb{R}^n$.

Exercise 4.37. Provide examples of linear subspaces V_1, V_2, V_3 of $\mathbb{R}^3$ satisfying

$$\{0\} \subsetneq V_1 \subsetneq V_2 \subsetneq V_3 \subseteq \mathbb{R}^3.$$

Exercise 4.38. Let e_1, e_2 denote the standard basis vectors of $\mathbb{R}^2$, that is,

$$e_1 := \begin{pmatrix} 1 \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Show that each of the following sets of vectors generates $\mathbb{R}^2$:

1. $\{e_1, e_2\}$,
2. $\{e_1, e_1 + e_2\}$,
3. $\{e_2, e_1 + e_2\}$,
4. $\{e_1 - e_2, e_1 + e_2\}$,
5. $\{e_1, e_2, e_1 + e_2\}$.

Prove that

$$\{ae_1 + be_2, ce_1 + de_2\}$$

generates $\mathbb{R}^2$ if and only if $ad - bc \neq 0$.

Exercise 4.39. Show that the linear subspace $\{0\}$ of $\mathbb{R}^n$ is generated by the set $\{0\}$.

Exercise 4.40. Determine a generating set for the subspace

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}.$$

Exercise 4.41. Determine a generating set for each of the subspaces as in the prior examples or exercises.

Exercise 4.42. Show that $\mathbb{R}^n$ is generated by the standard basis vectors $\{e_1, e_2, \dots, e_n\}$, where

$$e_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_3 := \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, e_n := \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

The following exercise shows that

$$\text{span}(S) + \text{span}(T) = \text{span}(S \cup T)$$

for any finite sets of vectors S and T in $\mathbb{R}^n$.

Exercise 4.43. Let U, V be linear subspaces of $\mathbb{R}^n$. Suppose U is generated by the set of vectors $\{u_1, u_2, \dots, u_s\}$ and V is generated by the set of vectors $\{v_1, v_2, \dots, v_t\}$. Show that the sum $U + V$ is generated by the set of vectors $\{u_1, u_2, \dots, u_s\} \cup \{v_1, v_2, \dots, v_t\}$.

Exercise 4.44. Show that the standard basis vectors

$$e_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_3 := \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, e_n := \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

form a basis of $\mathbb{R}^n$.

Exercise 4.45. Determine whether the subsets of $\mathbb{R}^2$ as in [Exercise 4.38](#) are bases of $\mathbb{R}^2$.

Exercise 4.46. Determine a basis of the linear subspace

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x - 2y + 6z = 0 \right\}.$$

of $\mathbb{R}^3$.

Exercise 4.47. Determine a basis of the linear subspace

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y = 0 \right\}.$$

of $\mathbb{R}^3$.

Exercise 4.48. Determine a basis of each of the subspaces as in the prior examples or exercises.

Exercise 4.49. Show that $\dim(\mathbb{R}^n) = n$.

Exercise 4.50. Determine a basis and the dimension of each of the subspaces as in the prior examples or exercises.

Exercise 4.51. Determine a basis and the dimension of the null space of

$$A = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -3 \end{pmatrix}.$$

Exercise 4.52. Determine a basis and the dimension of the column space of

$$A = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -3 \end{pmatrix}.$$

Chapter 5. Functions

Exercise 5.1. Consider the function $f : (0, 1) \rightarrow \mathbb{R}$ defined by

$$f(x) = \frac{1}{x} \text{ for all } x \in (0, 1).$$

Show that f is injective but not surjective.

Exercise 5.2. Let $M_n(\mathbb{R})$ denote the set of all $n \times n$ matrices with real entries. Show that the function $\det : M_3(\mathbb{R}) \rightarrow \mathbb{R}$, defined by

$$A \mapsto \det(A) \text{ for all } A \in M_3(\mathbb{R}),$$

is not injective, but surjective.

Exercise 5.3. Let $\text{GL}_2(\mathbb{R})$ denote the set of all invertible 2×2 matrices with real entries. Show that the function $\det : \text{GL}_2(\mathbb{R}) \rightarrow \mathbb{R} \setminus \{0\}$ defined by

$$A \mapsto \det(A) \text{ for all } A \in \text{GL}_2(\mathbb{R})$$

is surjective but not injective.

Exercise 5.4. Show that the map

$$f : \{\dots, -6, -4, -2, 0, 2, 4, 6, \dots\} \rightarrow \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

defined by

$$f(n) = \frac{n}{2} \text{ for all } n \in \{\dots, -6, -4, -2, 0, 2, 4, 6, \dots\}$$

is bijection between the set of all even integers and the set of all integers.

Exercise 5.5. Show that the map

$$f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$$

defined by

$$f(m, n) = 2^{m-1}(2n - 1) \text{ for all } (m, n) \in \mathbb{N} \times \mathbb{N}$$

is a bijection between $\mathbb{N} \times \mathbb{N}$ and $\mathbb{N}$.

Exercise 5.6. Let a, b be real numbers satisfying $a < b$. Show that the map

$$f : (0, 1) \rightarrow (a, b)$$

defined by

$$f(x) = a + (b - a)x \text{ for all } x \in (0, 1)$$

is a bijection.

Using the above or otherwise, show that for any real numbers a, b, c, d satisfying $a < b$ and $c < d$, there is a bijection between the intervals (a, b) and (c, d) .

Exercise 5.7. Let A be an $m \times n$ matrix with real entries. The **left multiplication by A** map

$$m_A : \mathbb{R}^n \rightarrow \mathbb{R}^m,$$

defined by

$$m_A(v) = Av \text{ for all } v \in \mathbb{R}^n$$

is

- injective if and only if the null space $\text{null}(A)$ is equal to $\{0\}$ of A ,

- surjective if and only if the column space $\text{col}(A)$ is equal to $\mathbb{R}^m$,
- bijective if and only if it is both injective and surjective, that is, if and only if $\text{null}(A) = \{0\}$ and $\text{col}(A) = \mathbb{R}^m$.

Exercise 5.8. Let S^1 be the unit circle in the plane, defined by

$$S^1 := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

Show that the function $f : \mathbb{R} \rightarrow S^1$ defined by

$$f(t) = \left(\frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2} \right) \text{ for all } t \in \mathbb{R}$$

is injective but not surjective.

Exercise 5.9. Let $\mathbb{R}[X]_{\leq d}$ be the set of all polynomials in one variable X with real coefficients and of degree at most d . Show that the function $f : \mathbb{R}^{d+1} \rightarrow \mathbb{R}[X]_{\leq d}$ defined by

$$f(a_0, a_1, \dots, a_d) = a_0 + a_1X + \dots + a_dX^d \text{ for all } (a_0, a_1, \dots, a_d) \in \mathbb{R}^{d+1}$$

is bijective.

Exercise 5.10. Let $\mathbb{R}[X]_{\leq d}$ be the set of all polynomials in one variable X with real coefficients and of degree at most d . Consider the map sending a polynomial to its even part, that is, the map $T_e : \mathbb{R}[X]_{\leq d} \rightarrow \mathbb{R}[X]_{\leq d}$ defined by

$$T_e(p(X)) = \frac{p(X) + p(-X)}{2} \text{ for all } p(X) \in \mathbb{R}[X]_{\leq d}.$$

Show that T_e is neither injective nor surjective.

Exercise 5.11. Let $\mathbb{R}[X]_{\leq d}$ be the set of all polynomials in one variable X with real coefficients and of degree at most d . Consider the map sending a polynomial to its odd part, that is, the map $T_o : \mathbb{R}[X]_{\leq d} \rightarrow \mathbb{R}[X]_{\leq d}$ defined by

$$T_o(p(X)) = \frac{p(X) - p(-X)}{2} \text{ for all } p(X) \in \mathbb{R}[X]_{\leq d}.$$

Show that T_o is neither injective nor surjective.

Exercise 5.12. The function

$$|\cdot| : \mathbb{R} \rightarrow \mathbb{R}$$

defined by

$$|x| := \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0 \end{cases}$$

is called the **absolute value** function. Show that it is neither injective nor surjective.

Exercise 5.13. Consider the function

$$f : \mathbb{R} \rightarrow (0, 1]$$

defined by

$$f(x) = \frac{1}{1 + |x|} \text{ for all } x \in \mathbb{R}.$$

Show that f is not injective. Determine $\text{im}(f)$.

Exercise 5.14. Let A be a nonempty set. Show that the identity map $\text{id}_A : A \rightarrow A$ is bijective.

Exercise 5.15. Let A be a nonempty subset of a set B . Show that the inclusion map $i : A \rightarrow B$ defined by

$$i(a) := a \text{ for all } a \in A$$

is injective. Is it surjective? Justify your answer.

Exercise 5.16. The map

$$\mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R},$$

defined by

$$(x, y) \mapsto |x - y| \text{ for all } (x, y) \in \mathbb{R} \times \mathbb{R},$$

is called the **distance function** on $\mathbb{R}$. Show that it is neither injective nor surjective.

Exercise 5.17. The map

$$d : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R},$$

defined by

$$d((x_1, y_1), (x_2, y_2)) := \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \text{ for all } (x_1, y_1), (x_2, y_2) \in \mathbb{R}^2,$$

is called the **distance function** on $\mathbb{R}^2$. Show that it is neither injective nor surjective.

Exercise 5.18. The map

$$\| \cdot \| : \mathbb{R}^2 \rightarrow \mathbb{R},$$

defined by

$$\|(x, y)\| := \sqrt{x^2 + y^2} \text{ for all } (x, y) \in \mathbb{R}^2,$$

is called the **norm** on $\mathbb{R}^2$. Show that it is neither injective nor surjective.

Exercise 5.19. Let u be an element of $\mathbb{R}^2$. Let θ be a real number, and let A_θ denote the orthogonal matrix

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Show that $\|A_\theta u\| = \|u\|$.

Exercise 5.20. The function

$$\mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R},$$

defined by

$$\langle (x_1, y_1), (x_2, y_2) \rangle := x_1 x_2 + y_1 y_2 \text{ for all } (x_1, y_1), (x_2, y_2) \in \mathbb{R}^2,$$

is called the **inner product** on $\mathbb{R}^2$. Show that the inner product of the standard basis vectors $e_1 = (1, 0)$ and $e_2 = (0, 1)$ is equal to 0.

Exercise 5.21. Does there exist a set S of three or more nonzero vectors in $\mathbb{R}^2$ such that the inner product of any two distinct vectors of S is equal to 0? Justify your answer.

Exercise 5.22. Let u, v be elements of $\mathbb{R}^2$ satisfying $\langle u, v \rangle = 0$. Let θ be a real number, and let A_θ denote the orthogonal matrix

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Show that the vectors $A_\theta u$ and $A_\theta v$ also satisfy

$$\langle A_\theta u, A_\theta v \rangle = 0.$$

Exercise 5.23. Show that the inner product on $\mathbb{R}^2$ has the following properties: for any $u, v, w \in \mathbb{R}^2$ and any element $c \in \mathbb{R}$, we have

- $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$,
- $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle$,
- $\langle cu, v \rangle = c\langle u, v \rangle$,
- $\langle u, cv \rangle = c\langle u, v \rangle$.

Exercise 5.24. For any u in $\mathbb{R}^2$, show that

$$\langle u, u \rangle = \|u\|^2.$$

Exercise 5.25. For any u, v in $\mathbb{R}^2$, show that

$$|\langle u, v \rangle| \leq \|u\| \|v\|.$$

Exercise 5.26. The map

$$\|\cdot\| : \mathbb{R}^3 \rightarrow \mathbb{R},$$

defined by

$$\|(x, y, z)\| := \sqrt{x^2 + y^2 + z^2} \text{ for all } (x, y, z) \in \mathbb{R}^3,$$

is called the **norm** on $\mathbb{R}^3$. Show that it is neither injective nor surjective.

Exercise 5.27. Show that the standard basis vectors of $\mathbb{R}^2$ satisfy $\|e_1\| = \|e_2\| = 1$.

Exercise 5.28. The map

$$d : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R},$$

defined by

$$d((x_1, y_1, z_1), (x_2, y_2, z_2)) := \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

$$\text{for all } (x_1, y_1, z_1), (x_2, y_2, z_2) \in \mathbb{R}^3,$$

is called the **distance function** on $\mathbb{R}^3$. Show that it is neither injective nor surjective.

Exercise 5.29. The function

$$\mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R},$$

defined by

$$\langle (x_1, y_1, z_1), (x_2, y_2, z_2) \rangle := x_1 x_2 + y_1 y_2 + z_1 z_2 \text{ for all } (x_1, y_1, z_1), (x_2, y_2, z_2) \in \mathbb{R}^3,$$

is called the **inner product** on $\mathbb{R}^3$. Show that the inner product of any two distinct standard basis vectors of $\mathbb{R}^3$ is equal to 0.

Exercise 5.30. Does there exist a set S of four or more nonzero vectors in $\mathbb{R}^3$ such that the inner product of any two distinct vectors of S is equal to 0? Justify your answer.

Exercise 5.31. Show that the inner product on $\mathbb{R}^3$ has the following properties: for any $u, v, w \in \mathbb{R}^3$ and any element $c \in \mathbb{R}$, we have

- $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$,
- $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle$,
- $\langle cu, v \rangle = c\langle u, v \rangle$,
- $\langle u, cv \rangle = c\langle u, v \rangle$.

Exercise 5.32. For any u in $\mathbb{R}^3$, show that

$$\langle u, u \rangle = \|u\|^2.$$

Exercise 5.33. Show that the standard basis vectors of $\mathbb{R}^3$ satisfy $\|e_1\| = \|e_2\| = \|e_3\| = 1$.

Exercise 5.34. Let A, B be nonempty sets. Let $f : A \rightarrow B$ be a function. Show that if f is invertible, then its inverse is unique.

Exercise 5.35. Let A, B, C be nonempty sets. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions.

- Show that if f and g are both injective (resp. surjective, bijective), then so is $g \circ f$.
- Show that a function $f : A \rightarrow B$ is bijective if it is invertible.

Exercise 5.36. Let A, B, C be nonempty sets. Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions.

- Show that if $g \circ f$ is injective, then so is f .
- Show that if $g \circ f$ is surjective, then so is g .

Exercise 5.37. Let $f : A \rightarrow B$ be a function. Show that

$$f(\emptyset) = \emptyset.$$

Exercise 5.38. Determine the images of the functions defined in the above exercises.

Exercise 5.39. Let $f : A \rightarrow B$ be a function. Show that for any two subsets $C_1, C_2 \subseteq A$, we have

- $f(C_1 \cup C_2) = f(C_1) \cup f(C_2)$,
- $f(C_1 \cap C_2) \subseteq f(C_1) \cap f(C_2)$.

What happens if f is injective?

Exercise 5.40. Let $f : A \rightarrow B$ be a function. Assume that for any subsets $C_1, C_2 \subseteq A$,

$$f(C_1 \cap C_2) = f(C_1) \cap f(C_2)$$

holds. Show that f is injective.

Note the the following exercise concludes the same result under a weaker assumption.

Exercise 5.41. Let $f : A \rightarrow B$ be a function. Assume that for any two subsets $C_1, C_2 \subseteq A$ satisfying $C_1 \cap C_2 = \emptyset$, we have

$$f(C_1) \cap f(C_2) = \emptyset.$$

Show that f is injective.

Exercise 5.42. Determine the inverse image of the set $\{1\}$ under the norm map

$$\|\cdot\| : \mathbb{R}^2 \rightarrow \mathbb{R}.$$

Exercise 5.43. Let $f : A \rightarrow B$ be an invertible function. Let $g : B \rightarrow A$ be its inverse. Show that for any subset $C \subseteq B$, the inverse image of C under f is equal to the image of C under g .

Exercise 5.44. Let $f : A \rightarrow B$ be a function. Show that for any two subsets $D_1, D_2 \subseteq B$,

- $f^{-1}(D_1 \cup D_2) = f^{-1}(D_1) \cup f^{-1}(D_2)$,
- $f^{-1}(D_1 \cap D_2) = f^{-1}(D_1) \cap f^{-1}(D_2)$.

Exercise 5.45. Let $f : A \rightarrow B$ be a function. For any subset $C \subseteq A$, show that

$$f^{-1}(f(C)) \supseteq C.$$

Exercise 5.46. Let $f : A \rightarrow B$ be a function. For any subset $D \subseteq B$, show that

$$f(f^{-1}(D)) \subseteq D.$$

Exercise 5.47. Let $f : A \rightarrow B$ be a function. Show that the following statements are equivalent:

- f is surjective.
- For any subset $D \subseteq B$, we have $f(f^{-1}(D)) = D$.
- For any subset $C \subseteq A$, we have $f(A \setminus C) \supseteq B \setminus f(C)$.

Exercise 5.48. Let $f : A \rightarrow B$ be a function. Show that the following statements are equivalent:

- f is injective.
- For any subset $D \subseteq B$, we have $f^{-1}(f(D)) = D$.
- For any subset $C \subseteq A$, we have $f(A \setminus C) \subseteq B \setminus f(C)$.

Exercise 5.49. Let $f : A \rightarrow B$ be an invertible function. Show that for any subset $C \subseteq A$ and any subset $D \subseteq B$, we have

- $f^{-1}(f(C)) = C$,
- $f(f^{-1}(D)) = D$.

Exercise 5.50. Let $f : A \rightarrow B$ be a function. For any subset $C \subseteq A$, show that $C \subseteq f^{-1}(f(C))$. What happens if f is injective?

Practice Problems

Chapter 6. Sets

Exercise 6.1. Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$. Find

$$A \cap B, A \cup B, A \setminus B, B \setminus A, A \Delta B.$$

Exercise 6.2. Let $U = \{1, 2, \dots, 10\}$ be the universal set. If $A = \{2, 4, 6, 8, 10\}$ and $B = \{1, 3, 5, 7, 9\}$, then find

$$A^c, B^c, (A \cup B)^c, (A \cap B)^c.$$

Exercise 6.3. Write down the power set of each of the following sets:

$$\{1, 2\}, \{1, 2, 3\}, \{1, 2, 3, 4\}.$$

Exercise 6.4. Identify the set

$$\left\{x \in \mathbb{R} \setminus \{0\} : x + \frac{1}{x} \geq 2\right\}.$$

Exercise 6.5. Let A, B be subsets of $\mathbb{R}$ defined by

$$A = \{x \in \mathbb{R} : x^2 \geq 0\},$$

$$B = \{x \in \mathbb{R} : x^3 \geq 0\}.$$

Determine the sets $A \cup B, A \cap B, A \setminus B, B \setminus A$.

Exercise 6.6. For a positive integer k , let A_k denote the set of integral multiples of k , that is,

$$A_k := \{r \in \mathbb{Z} : r = k\ell \text{ for some } \ell \in \mathbb{Z}\}.$$

Let m, n be positive integers. For each of the following statements, determine the equivalent conditions on the integers m, n .

1. $A_m \subseteq A_n$
2. $A_m \subsetneq A_n$
3. $A_m \not\subseteq A_n$
4. $A_m = A_n$

Chapter 7. Induction

Exercise 7.1. Show that

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

for all $n \in \mathbb{N}$.

Exercise 7.2. Show that

$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \cdots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$

for all $n \in \mathbb{N}$.

Exercise 7.3. Show that

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{2 \cdot 3 \cdot 4} + \cdots + \frac{1}{n(n+1)(n+2)} = \frac{n(n+3)}{4(n+1)(n+2)}$$

for all $n \in \mathbb{N}$.

Exercise 7.4. Show that

$$3 + 11 + \cdots + (8n - 5) = 4n^2 - n$$

for all $n \in \mathbb{N}$.

Exercise 7.5. Show that

$$1^2 + 3^2 + \cdots + (2n-1)^2 = \frac{4n^3 - n}{3}$$

for all $n \in \mathbb{N}$.

Exercise 7.6. Show that

$$1^2 - 2^2 + 3^2 - 4^2 + \cdots + (-1)^{n+1}n^2 = (-1)^{n+1} \frac{n(n+1)}{2}$$

for all $n \in \mathbb{N}$.

Exercise 7.7. Show that

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6},$$

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

hold for any positive integer n .

Exercise 7.8. Show that

$$\frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{2n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{2n-1} - \frac{1}{2n}$$

for all $n \in \mathbb{N}$.

Exercise 7.9. Show that $n^3 + 5n$ is divisible by 6 for all $n \in \mathbb{N}$.

Exercise 7.10. Show that $5^{2n} - 1$ is divisible by 8 for all $n \in \mathbb{N}$.

Exercise 7.11. Show that $5^n - 4n - 1$ is divisible by 16 for all $n \in \mathbb{N}$.

Exercise 7.12. Show that $6^n - 5n - 1$ is divisible by 25 for all $n \in \mathbb{N}$.

Exercise 7.13. Show that $n^3 + (n+1)^3 + (n+2)^3$ is divisible by 9 for all $n \in \mathbb{N}$.

Exercise 7.14. Show that

$$\frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \cdots + \frac{1}{\sqrt{n}} > \sqrt{n}$$

for all $n \in \mathbb{N}$ satisfying $n > 1$.

Chapter 8. Matrices

Exercise 8.1. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & 1 \\ 1 & 6 \end{pmatrix}$. Compute $2A + 3B$, A^2B , AB^3 , A^T .

Exercise 8.2. Suppose A is a 3×2 matrix, and

$$A \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 8 \\ 14 \end{pmatrix},$$

$$A \begin{pmatrix} 3 \\ 7 \end{pmatrix} = \begin{pmatrix} -4 \\ 8 \\ 20 \end{pmatrix}.$$

Determine

$$A \begin{pmatrix} 103 \\ 407 \end{pmatrix}, A \begin{pmatrix} 97 \\ 393 \end{pmatrix}.$$

Exercise 8.3. Show that every vector in $\mathbb{R}^2$ can be expressed as a linear combination of the vectors

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix}.$$

Exercise 8.4. Write the following system of equations in matrix form $Ax = b$.

$$2x + 3y = 5,$$

$$4x - y = 1.$$

Determine if A is invertible. If it is, find A^{-1} and use it to solve the system.

Exercise 8.5. If A, B are 2×2 matrices, then show that

$$\det(AB) = \det(A) \det(B).$$

Exercise 8.6. Determine which of the following matrices are upper-triangular, lower-triangular, or neither.

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}, \begin{pmatrix} 7 & 8 \\ 0 & 9 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{pmatrix}, \begin{pmatrix} 7 & 0 \\ 8 & 9 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 4 \\ 5 & 0 & 6 \end{pmatrix}.$$

Exercise 8.7. Determine which of the following matrices are symmetric, which are skew-symmetric, and which are neither.

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}, \begin{pmatrix} 7 & 8 \\ 8 & 9 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix},$$

$$\begin{pmatrix} 0 & 2 & 3 \\ -2 & 0 & 5 \\ -3 & -5 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 8 \\ -8 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Exercise 8.8. Determine which of the following matrices are orthogonal.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

Exercise 8.9. Let

$$A = \begin{pmatrix} 1 & -5 & 6 \\ -4 & 8 & -9 \\ 7 & 2 & 10 \end{pmatrix}.$$

Find $\det(A)$ and $\text{adj}(A)$.

Exercise 8.10. Let

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 3 & 2 & 1 \\ 4 & 1 & 2 \end{pmatrix}.$$

Find $\det(A)$ and $\text{adj}(A)$.

Exercise 8.11. Let θ be a real number satisfying $0 < \theta < \frac{\pi}{2}$. Show that the matrix

$$A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

is orthogonal.

Exercise 8.12. Let A be an orthogonal matrix. Show that A is invertible and that

$$A^{-1} = A^T.$$

Exercise 8.13. Write the following system of equations in matrix form $Ax = b$.

$$\begin{aligned} x_1 + 2x_2 - x_3 + 4x_4 &= 5, \\ 2x_1 + 3x_2 + x_3 + 5x_4 &= 8, \\ -x_1 + 4x_2 - 2x_3 - 3x_4 &= -4, \\ 3x_1 - x_2 + 4x_3 + 2x_4 &= 7. \end{aligned}$$

Perform Gaussian elimination on the augmented matrix $(A \mid b)$, to reduce it to row echelon form. Using this form, determine if the system has a solution. If it does, find all solutions of the above system in the variables x_1, x_2, x_3, x_4 .

Exercise 8.14. Perform row reduction on the matrix

$$\begin{pmatrix} 1 & 2 & -5 \\ 2 & 5 & -7 \\ -1 & -2 & 4 \end{pmatrix}$$

to reduce it to row echelon form. Using these reductions, find the inverse of the above matrix, if it exists.

Exercise 8.15. Transform the matrix

$$\begin{pmatrix} 0 & 2 & -3 & 0 & 0 \\ 1 & -1 & 4 & 0 & 5 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 3 & -5 & 0 & 1 \\ 0 & 3 & -5 & -1 & 2 \end{pmatrix}$$

into a matrix in row echelon form using elementary row operations.

Exercise 8.16. Solve the system of equations in the variables x_1, x_2, x_3, x_4 .

$$\begin{aligned} 2x_2 - 3x_3 &= 0, \\ x_1 - x_2 + 4x_3 &= 5, \\ x_4 &= -2, \\ 3x_2 - 5x_3 &= 1, \\ 3x_2 - 5x_3 - x_4 &= 2. \end{aligned}$$

Exercise 8.17. Let

$$A = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix},$$

where a, b, c are scalars. Show that $\det(A) = abc$, and that

$$\operatorname{adj}(A) = \begin{pmatrix} bc & 0 & 0 \\ 0 & ca & 0 \\ 0 & 0 & ab \end{pmatrix}.$$

Exercise 8.18. Compute the determinant of the matrix

$$A = \begin{pmatrix} 1 & -5 & 6 \\ 2 & 0 & 9 \\ -1 & 2 & -1 \end{pmatrix}$$

to determine whether it is invertible. If A is invertible, then find its inverse using its adjoint.

Exercise 8.19. Use Gaussian elimination to determine the row echelon form of the matrix

$$A = \begin{pmatrix} 1 & -5 & 6 \\ 2 & 0 & 9 \\ -1 & 2 & -1 \end{pmatrix}.$$

Use the row echelon form to determine whether A is invertible. If A is invertible, then find its inverse using elementary matrices, corresponding to the elementary row operations used to transform A into its row echelon form.

Exercise 8.20. Solve the following system of equations in the variables x, y, z .

$$\begin{aligned} x + 2y + 3z &= 1, \\ 4x - 5y + 6z &= 2, \\ 7x - y + 10z &= 3. \end{aligned}$$

Chapter 9. The space $\mathbb{R}^n$

Exercise 9.1. Let S be a subset of $\mathbb{R}^n$, containing precisely one element v . Show that S is linearly independent if and only if $v \neq 0$.

Exercise 9.2. In each of the following, determine whether the given vectors are linearly independent or dependent. Justify your answers.

1. $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$, and $\begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}$ in $\mathbb{R}^3$.
2. $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, and $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ in $\mathbb{R}^3$.
3. $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 4 \end{pmatrix}$ in $\mathbb{R}^2$.
4. $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ in $\mathbb{R}^2$.
5. $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$, and $\begin{pmatrix} 7 \\ 8 \\ 10 \end{pmatrix}$ in $\mathbb{R}^3$.

Exercise 9.3. Let $v_1 = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$, $v_2 = \begin{pmatrix} 4 \\ 0 \\ -1 \end{pmatrix}$, $v_3 = \begin{pmatrix} -1 \\ 5 \\ 2 \end{pmatrix}$. Determine whether the vectors v_1, v_2, v_3 are linearly independent or dependent. Justify your answer.

Exercise 9.4. Determine whether the vectors $\begin{pmatrix} 1 \\ 2 \\ -1 \\ 3 \end{pmatrix}$, $\begin{pmatrix} 4 \\ 0 \\ 5 \\ -2 \end{pmatrix}$, $\begin{pmatrix} -2 \\ 1 \\ 3 \\ 4 \end{pmatrix}$, and $\begin{pmatrix} 3 \\ -1 \\ 2 \\ 1 \end{pmatrix}$ in $\mathbb{R}^4$ are linearly independent or dependent. Justify your answer.

Exercise 9.5. Show that the standard basis vectors

$$e_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_3 := \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, e_n := \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

of $\mathbb{R}^n$ are linearly independent.

Exercise 9.6. Show that any finite subset of $\mathbb{R}^n$ containing the zero vector is linearly dependent.

Exercise 9.7. Let V denote the set of all vectors of $\mathbb{R}^n$ whose first component is equal to zero, that is,

$$V = \left\{ \begin{pmatrix} 0 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix} : x_i \in \mathbb{R} \text{ for all } i = 2, 3, \dots, n \right\}.$$

Show that V is a linear subspace of $\mathbb{R}^n$.

Exercise 9.8. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}$ is a linear subspace of $\mathbb{R}^3$.

Exercise 9.9. Let (a, b, c) be a nonzero element of $\mathbb{R}^3$. Show that the set

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : ax + by + cz = 0 \right\}$$

is a linear subspace of $\mathbb{R}^3$.

Exercise 9.10. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x \geq 0, y \geq 0 \right\}$ is not a linear subspace of $\mathbb{R}^2$.

Exercise 9.11. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x = 1 \right\}$ is not a linear subspace of $\mathbb{R}^2$.

Exercise 9.12. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : x = y \right\}$ is a linear subspace of $\mathbb{R}^2$.

Exercise 9.13. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : z = 0 \right\}$ is a linear subspace of $\mathbb{R}^3$.

Exercise 9.14. Show that the set $V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 1 \right\}$ is not a linear subspace of $\mathbb{R}^3$.

Exercise 9.15. Let A be an $m \times n$ matrix. Let V be the set of solutions to the system of equations

$$Au = 0,$$

that is,

$$V = \{v \in \mathbb{R}^n : Av = 0\}.$$

Show that V is a linear subspace of $\mathbb{R}^n$.

Exercise 9.16. Let A be an $m \times n$ matrix. Let V be the set of vectors which can be expressed as a linear combination of the columns of A , that is,

$$V = \{v \in \mathbb{R}^m : v = Au \text{ for some } u \in \mathbb{R}^n\}.$$

Show that V is a linear subspace of $\mathbb{R}^m$.

Exercise 9.17. Let U, V be the subsets of $\mathbb{R}^3$ defined as

$$U = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + 2y = 0 \right\},$$

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}.$$

Show that both U and V are linear subspaces of $\mathbb{R}^3$. Show that $U \cap V$ is a linear subspace of $\mathbb{R}^3$. Determine a basis for the intersection $U \cap V$ of U and V .

Exercise 9.18. Let V be a linear subspace of $\mathbb{R}^n$. Show that $V + V = V$.

Exercise 9.19. Let U, V be the subsets of $\mathbb{R}^3$ defined as

$$U = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + 2y = 0 \right\},$$

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}.$$

Determine the sum of U and V .

Exercise 9.20. Provide examples of linear subspaces V_1, V_2, V_3 of $\mathbb{R}^3$ satisfying $\{0\} \subsetneq V_1 \subsetneq V_2 \subsetneq V_3 \subseteq \mathbb{R}^3$.

Exercise 9.21. Let e_1, e_2 denote the standard basis vectors of $\mathbb{R}^2$, that is,

$$e_1 := \begin{pmatrix} 1 \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Show that each of the following sets of vectors generates $\mathbb{R}^2$:

1. $\{e_1, e_2\}$,
2. $\{e_1, e_1 + e_2\}$,
3. $\{e_2, e_1 + e_2\}$,
4. $\{e_1 - e_2, e_1 + e_2\}$,
5. $\{e_1, e_2, e_1 + e_2\}$.

Prove that

$$\{ae_1 + be_2, ce_1 + de_2\}$$

generates $\mathbb{R}^2$ if and only if $ad - bc \neq 0$.

Exercise 9.22. Determine a generating set for the subspace

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}.$$

Exercise 9.23. Show that $\mathbb{R}^n$ is generated by the standard basis vectors $\{e_1, e_2, \dots, e_n\}$, where

$$e_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_3 := \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, e_n := \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

Exercise 9.24. Show that the standard basis vectors

$$e_1 := \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_2 := \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, e_3 := \begin{pmatrix} 0 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, e_n := \begin{pmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$$

form a basis of $\mathbb{R}^n$.

Exercise 9.25. Determine whether the subsets of $\mathbb{R}^2$ as in [Exercise 9.21](#) are bases of $\mathbb{R}^2$.

Exercise 9.26. Determine a basis of the linear subspace

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x - 2y + 6z = 0 \right\}.$$

of $\mathbb{R}^3$.

Exercise 9.27. Determine a basis of the linear subspace

$$V = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3 : x + y = 0 \right\}.$$

of $\mathbb{R}^3$.

Exercise 9.28. Determine a basis of each of the subspaces as in the prior examples or exercises.

Exercise 9.29. Show that $\dim(\mathbb{R}^n) = n$.

Exercise 9.30. Determine a basis and the dimension of each of the subspaces as in the prior examples or exercises.

Exercise 9.31. Determine a basis and the dimension of the null space of

$$A = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -3 \end{pmatrix}.$$

Exercise 9.32. Determine a basis and the dimension of the column space of

$$A = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -3 \end{pmatrix}.$$

Chapter 10. Functions

Exercise 10.1. Consider the function $f : (0, 1) \rightarrow \mathbb{R}$ defined by

$$f(x) = \frac{1}{x} \text{ for all } x \in (0, 1).$$

Show that f is injective but not surjective.

Exercise 10.2. Let $M_n(\mathbb{R})$ denote the set of all $n \times n$ matrices with real entries. Show that the function $\det : M_3(\mathbb{R}) \rightarrow \mathbb{R}$, defined by

$$A \mapsto \det(A) \text{ for all } A \in M_3(\mathbb{R}),$$

is not injective, but surjective.

Exercise 10.3. Let $\text{GL}_2(\mathbb{R})$ denote the set of all invertible 2×2 matrices with real entries. Show that the function $\det : \text{GL}_2(\mathbb{R}) \rightarrow \mathbb{R} \setminus \{0\}$ defined by

$$A \mapsto \det(A) \text{ for all } A \in \text{GL}_2(\mathbb{R})$$

is surjective but not injective.

Exercise 10.4. Show that the map

$$f : \{\dots, -6, -4, -2, 0, 2, 4, 6, \dots\} \rightarrow \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$$

defined by

$$f(n) = \frac{n}{2} \text{ for all } n \in \{\dots, -6, -4, -2, 0, 2, 4, 6, \dots\}$$

is a bijection between the set of all even integers and the set of all integers.

Exercise 10.5. Show that the map

$$f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$$

defined by

$$f(m, n) = 2^{m-1}(2n - 1) \text{ for all } (m, n) \in \mathbb{N} \times \mathbb{N}$$

is a bijection between $\mathbb{N} \times \mathbb{N}$ and $\mathbb{N}$.

Exercise 10.6. Let a, b be real numbers satisfying $a < b$. Show that the map

$$f : (0, 1) \rightarrow (a, b)$$

defined by

$$f(x) = a + (b - a)x \text{ for all } x \in (0, 1)$$

is a bijection.

Using the above or otherwise, show that for any real numbers a, b, c, d satisfying $a < b$ and $c < d$, there is a bijection between the intervals (a, b) and (c, d) .

Exercise 10.7. The function

$$|\cdot| : \mathbb{R} \rightarrow \mathbb{R}$$

defined by

$$|x| := \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0 \end{cases}$$

is called the **absolute value** function. Show that it is neither injective nor surjective.

Exercise 10.8. Consider the function

$$f : \mathbb{R} \rightarrow (0, 1]$$

defined by

$$f(x) = \frac{1}{1 + |x|} \text{ for all } x \in \mathbb{R}.$$

Show that f is not injective. Determine $\text{im}(f)$.

Exercise 10.9. The map

$$\|\cdot\| : \mathbb{R}^2 \rightarrow \mathbb{R},$$

defined by

$$\|(x, y)\| := \sqrt{x^2 + y^2} \text{ for all } (x, y) \in \mathbb{R}^2,$$

is called the **norm** on $\mathbb{R}^2$. Show that it is neither injective nor surjective.

Exercise 10.10. Determine the inverse image of the set $\{1\}$ under the norm map

$$\|\cdot\| : \mathbb{R}^2 \rightarrow \mathbb{R}.$$